EXPERIMENT 14: Install a C Compiler in a Virtual Machine and Execute a File Read/Write Program

1. Aim

To install a C compiler in a virtual machine and execute a C program to perform file write and read operations.

2. Procedure

1. Open Oracle VirtualBox or VMware Workstation.
2. Start a virtual machine with an installed Linux operating system.
3. Open the Terminal application.
4. Update the package repository using sudo apt update.
5. Install GCC using sudo apt install gcc -y.
6. Verify the compiler installation using gcc --version.
7. Create a source file named fileoperation.c using a text editor.
8. Enter the C program to write data into a file and read it back.
9. Save the file and compile it using GCC.
10. Execute the compiled program.
11. Enter the text to be written into the file.
12. Verify the success message displayed after the write operation.
13. Observe the file contents displayed by the read operation.
14. Use ls or a file manager to confirm that sample.txt has been created.

3. Program

Install GCC:

```
sudo apt update
sudo apt install gcc -y
gcc --version
```

Program – fileoperation.c:

```
#include <stdio.h>

int main()
{
    FILE *fp;
    char text[100];

    // Write operation
    fp = fopen("sample.txt", "w");
    if (fp == NULL) {
        printf("File cannot be opened");
        return 1;
    }

    printf("Enter text to write into file: ");
    fgets(text, sizeof(text), stdin);
    fprintf(fp, "%s", text);
    fclose(fp);

    printf("\nData written successfully.\n");

    // Read operation
    fp = fopen("sample.txt", "r");
    if (fp == NULL) {
```

```
        printf("File cannot be opened");
        return 1;
    }

    printf("\nContents of the file:\n");
    while (fgets(text, sizeof(text), fp) != NULL) {
        printf("%s", text);
    }

    fclose(fp);
    return 0;
}
```

Compile and execute:

```
gcc fileoperation.c -o fileoperation
./fileoperation
```

4. Output of Program

```
Enter text to write into file:
Cloud Computing Lab

Data written successfully.

Contents of the file:
Cloud Computing Lab

Created file: sample.txt
```

5. Result

Thus, GCC was installed in the virtual machine, and the C program to perform file read and write operations was compiled and executed successfully.

EXPERIMENT 14: Install a C Compiler in a Virtual Machine and Execute a File Read/Write Program

1. Aim

To install and configure the GCC compiler in a Linux-based virtual machine and develop, compile and execute a C program to perform file write and read operations using standard file handling functions.

2. Procedure

1. Start the host computer and open Oracle VirtualBox or VMware Workstation. 2. Start a virtual machine with an installed Linux operating system. 3. Log in to the Linux virtual machine using the configured user account. 4. Open the Terminal application to execute the required Linux commands. 5. Update the package repository using sudo apt update. 6. Install the GCC compiler using sudo apt install gcc -y. 7. Verify the compiler installation using gcc --version. 8. Create a C source file named fileoperation.c using a text editor. 9. Enter the C program to perform file writing and reading operations. 10. Include the required header file and declare a file pointer using the FILE data type. 11. Use the fopen() function with write mode to create or open the file sample.txt. 12. Use fgets() to read the input text from the user and fprintf() to write the data into the file. 13. Close the file using fclose() to ensure that the written data is properly saved. 14. Reopen the same file using fopen() in read mode. 15. Use fgets() inside a loop to read the file contents. 16. Display the retrieved contents on the terminal using printf(). 17. Close the file after completing the read operation. 18. Save the C source file and compile it using gcc fileoperation.c -o fileoperation. 19. Execute the generated program using ./fileoperation. 20. Enter a sample text when prompted and observe the write confirmation message. 21. Verify that the contents written into the file are displayed correctly. 22. Use the ls command to confirm that sample.txt has been created in the current directory.

4. Output of Program

Sample Input: Enter text to write into file: Cloud Computing Lab Expected Output: Data written successfully. Contents of the file: Cloud Computing Lab Created file: sample.txt Explanation: The program performs two major file operations: - Write Operation: The fopen() function opens the file in write mode, and fprintf() stores the entered data. - Read Operation: The file is reopened in read mode, and fgets() retrieves the stored contents for display. The fclose() function is used to close the file after each operation.

5. Result

Thus, GCC was successfully installed and configured in the Linux virtual machine. The C program was compiled and executed successfully to perform file write and read operations, and the stored contents were displayed correctly.